

How local Tate duality is proved

The purely local fact — what it rests on, why it already contains local class field theory, and the two instances the rest of these notes use

on a question of Rene Pannekoek; checked in `local-duality.gp`, output in `results/local-duality.txt`

The short answer. Everything reduces to one computation, $\text{inv} : \text{Br}(K) \cong \mathbb{Q}/\mathbb{Z}$, and then to formal cohomological machinery. But the theorem is not **cheaper** than local class field theory: the case $M = \mathbb{Z}/n$, $i = 1$ of the duality statement **is** the norm residue isomorphism. There is no route to the local fact that bypasses LCFT — they sit at the same depth. What is genuinely soft is the **tame** part: away from the residue characteristic the unramified subgroups are exact annihilators for elementary reasons, and that is the version the survey actually uses at good places.

1. The statement

Let $K/\mathbb{Q}_p$ be finite, M a finite G_K -module, and $M^D = \text{Hom}(M, \overline{K}^\times)$ its Cartier dual. Cup product followed by the invariant gives

$$H^i(K, M) \times H^{2-i}(K, M^D) \longrightarrow H^2(K, \overline{K}^\times) = \text{Br}(K) \xrightarrow{\text{inv}} \mathbb{Q}/\mathbb{Z},$$

and the theorem is that this is a **perfect pairing of finite groups** for $i = 0, 1, 2$.

For abelian varieties there is the companion statement

$$A(K)^\wedge \times H^1(K, \hat{A}) \longrightarrow \mathbb{Q}/\mathbb{Z},$$

perfect, with $A(K)$ profinitely completed and H^1 discrete torsion. This is the pairing that `ec-density-bm.typ` writes as $\langle Q_v, \beta_v \rangle_v$. It follows from the finite-module case by a limit argument (Section 8).

2. The engine: $\text{inv} : \text{Br}(K) \cong \mathbb{Q}/\mathbb{Z}$

This is the one hard local input. It comes in two halves: **every class is split by an unramified extension**, and **for unramified cyclic extensions the group can be computed by hand**. Together they give $\text{Br}(K) = \bigcup_n \text{Br}(K_n/K) = \bigcup_n \frac{1}{n} \mathbb{Z}/\mathbb{Z}$.

2.1. A valuation on a division algebra, and why uniqueness is the point

Let D be a central division algebra over K of degree n , so $\dim_K D = n^2$. One wants to do to D what one does to a field: put a valuation on it, take its valuation ring, and look at the residue algebra. The obstacle is that D is not commutative, so there is no direct construction.

The way in is that **every single element of D lives in a field**: for $x \in D$, the subring $K(x)$ generated by K and x is commutative, hence a finite field extension of K . So one can try to define

$$w(x) := \text{the extension of } v \text{ to } K(x), \text{ evaluated at } x.$$

This is exactly where uniqueness is used, and it is used twice.

(1) **Well-definedness.** K is complete, so v extends to a finite extension F/K in **exactly one** way, by $v_F(y) = \frac{1}{[F:K]} \cdot v(N_{F/K}(y))$. Were there several extensions, $w(x)$ would depend on a choice of extension — and worse, on which subfield of D one chose to view x in, since a given x lies in many. Uniqueness makes w a well-defined function on all of D at once, and makes the values computed in different subfields automatically consistent.

(2) Checking the axioms. $w(xy) = w(x) + w(y)$ is multiplicativity of the reduced norm. The ultrametric inequality is the one that has no non-commutative proof — but it does not need one: dividing through, it reduces to $w(1+z) \geq \min(0, w(z))$, and $1+z$ lies in the **field** $K(z)$, where the inequality is the ordinary one. Uniqueness is what lets the value computed inside $K(z)$ be the value in D .

So w is a valuation, $\mathcal{O}_D = \{w \geq 0\}$ is a ring — the unique maximal order — with a unique maximal ideal $\mathfrak{m}_D = \{w > 0\}$, and one may form the **residue algebra**

$$\overline{D} = \mathcal{O}_D / \mathfrak{m}_D,$$

a finite-dimensional division algebra over the finite residue field k of K . Without the uniqueness there is no w , hence no $\mathcal{O}_D$, no $\overline{D}$, and the whole route below collapses at the first step.

2.2. The residue algebra is a field, and $e = f = n$

Two facts now do the work.

- **Wedderburn’s little theorem:** a finite division ring is commutative. So $\overline{D}$ is a **field**, a finite extension of k ; write $f = [\overline{D} : k]$.
- **The degree formula** $\dim_K D = ef$, where $e = [w(D^\times) : v(K^\times)]$ — the same $[L : K] = ef$ that holds for field extensions, and proved the same way, by lifting a k -basis of $\overline{D}$ and multiplying by powers of a uniformiser of D .

Since w takes values in $\frac{1}{n}\mathbb{Z}$ (it is $\frac{1}{n}$ times a valuation of K , via the reduced norm) one has $e \leq n$; and $f \leq n$ because $\overline{D}$ embeds in a maximal subfield’s residue field. But $ef = \dim_K D = n^2$. Two numbers at most n with product n^2 must both equal n :

$$e = f = n.$$

In particular $\overline{D}/k$ has degree exactly n — the residue algebra is as large as it could possibly be.

2.3. Lifting, and why a maximal subfield splits

$\overline{D}/k$ is an extension of finite fields, hence generated by one element $\overline{\alpha}$. Hensel lifts $\overline{\alpha}$ to $\alpha \in \mathcal{O}_D$ with $K(\alpha)/K$ **unramified of degree n** . Since the maximal subfields of D have degree exactly n , $K(\alpha)$ is a maximal subfield, and it is unramified.

It remains that a maximal subfield splits:

If $L \subset D$ is a maximal subfield then $D \otimes_K L \cong M_n(L)$. Let D act on itself by left multiplication and let L act by right multiplication; the two commute, so $D \otimes_K L \rightarrow \text{End}_L(D)$. D is an L -vector space of dimension n , so the target is $M_n(L)$. The source is a simple algebra, so the map is injective; and both sides have L -dimension n^2 . Injective plus equal dimensions gives an isomorphism. ■

So $[D]$ dies in $\text{Br}(L)$ for an unramified L/K , which is the first half: $\text{Br}(K) = \bigcup_n \text{Br}(K_n/K)$, K_n the unramified extension of degree n .

2.4. Computing $\text{Br}(L/K)$ for L unramified

Let L/K be unramified cyclic of degree n , $G = \text{Gal}(L/K) = \langle \sigma \rangle$. Two steps.

Step 1: $\text{Br}(L/K) \cong H^2(G, L^\times)$. This is the crossed-product theorem — a central simple algebra split by L is a crossed product $\oplus_{\tau \in G} Lu_\tau$, and associativity of its multiplication table is the 2-cocycle condition. It is written out in §6 of `csa-brauer.typ`, so nothing new is needed here.

Step 2: for G cyclic, $H^2(G, M) \cong M^G / N_G(M)$, so $H^2(G, L^\times) = K^\times / N_{L/K}(L^\times)$. Now compute that norm group, and this is where “unramified” enters:

- $L^\times = \pi^\mathbb{Z} \times \mathcal{O}_L^\times$, with π a uniformiser of K — and π stays a uniformiser of L precisely because L/K is unramified. So $N(\pi) = \pi^n$.
- **Units are norms:** $N(\mathcal{O}_L^\times) = \mathcal{O}_K^\times$. On residue fields the norm $\ell^\times \rightarrow k^\times$ is $x \mapsto x^{(q^f-1)/(q-1)}$, whose image is the subgroup of index $(q^f-1)/(q-1)$ in a cyclic group of order q^f-1 — that is, all of $k^\times$. Then one climbs the unit filtration: on $U^{(i)}/U^{(i+1)}$ the norm becomes the **trace** $\ell \rightarrow k$, surjective because the extension is separable. Completeness assembles the successive approximations.

Hence $N(L^\times) = \{x \in K^\times : n \mid v(x)\}$, and

$$K^\times/N(L^\times) \cong \mathbb{Z}/n, \quad x \mapsto v(x) \bmod n, \quad \text{inv}(x) := v(x)/n \in \frac{1}{n}\mathbb{Z}/\mathbb{Z}.$$

One check remains: that these invariants are **compatible** as n grows, so that $\frac{1}{n}\mathbb{Z}/\mathbb{Z} \hookrightarrow \frac{1}{nm}\mathbb{Z}/\mathbb{Z}$ under inflation $K_n \subset K_{nm}$. That is a direct computation with norms and Frobenius, and in the limit gives $\text{inv} : \text{Br}(K) \cong \mathbb{Q}/\mathbb{Z}$.

2.5. The $n = 2$ case, checked

Everything above is visible in the quaternion case, where the Hilbert symbol computes it. Take u a non-residue unit, so $K = \mathbb{Q}_p(\sqrt{u})$ is the unramified quadratic extension.

statement	check
units are norms	$(u, w)_p = +1$ for every unit w , at every odd $p < 40$ — no exceptions
a uniformiser is not	$(u, p)_p = -1$ at every such p
so $N(L^\times) = \{v \text{ even}\}$ and $\text{inv}(x) = v(x)/2$	the index is 2, detected by $v_p \bmod 2$
the unramified subfield sits in the division algebra and splits it	(u, p) is a division algebra since $(u, p)_p = -1$, and u becomes a square in $\mathbb{Q}_p(\text{sqrt } u)$, which therefore splits it

and the residue-field input of Step 2 is checked directly: the norm $\mathbb{F}_{q^f}^\times \rightarrow \mathbb{F}_q^\times$ is surjective for every prime $q \leq 19$ and $f \leq 4$.

3. Why the theorem contains local class field theory

Take $M = \mathbb{Z}/n$ with trivial action, so $M^D = \mu_n$, and look at $i = 1$:

$$H^1(K, \mathbb{Z}/n) \times H^1(K, \mu_n) = \text{Hom}(G_K, \mathbb{Z}/n) \times K^\times / (K^\times)^n \longrightarrow \frac{1}{n}\mathbb{Z}/\mathbb{Z}, \quad (\chi, a) \mapsto \chi(\text{rec}(a)).$$

Perfectness of **this** pairing is precisely the statement that the reciprocity map $\text{rec} : K^\times \rightarrow G_K^{\text{ab}}$ induces an isomorphism $K^\times / (K^\times)^n \cong G_K^{\text{ab}}/n$ — the norm residue isomorphism. So local duality is not a lemma on the way to LCFT; it has LCFT inside it. The cases $i = 0, 2$ for the same M are just inv again.

3.1. The instance you can check by hand: $n = 2$

$\mu_2 = \{\pm 1\}$ already lies in $\mathbb{Q}_p$, so both sides are $\mathbb{Q}_p^\times / (\mathbb{Q}_p^\times)^2$ and the pairing is the Hilbert symbol. Perfectness is then literally non-degeneracy of the Gram matrix over $\mathbb{F}_2$:

p	$\#(\mathbb{Q}_p^\times / \text{sq})$	Gram rank over $\mathbb{F}_2$	radical
2	8	3 of 3	trivial
3, 5, 7, 11, 13	4	2 of 2	trivial

At every p , including 2, the only class pairing trivially with everything is the trivial one. That is local Tate duality for $\mathbb{Z}/2$, verified exhaustively.

4. The proof proper: dévissage

With the $M = \mathbb{Z}/n$ case in hand, the rest is formal. Write $\alpha^i(M) : H^i(K, M) \rightarrow H^{2-i}(K, M^D)^\vee$ for the map induced by the pairing. One shows α^i is an isomorphism by:

1. **Finiteness.** $H^i(K, M)$ is finite for finite M — so Pontryagin duality behaves and one may argue by counting.
2. **The five lemma.** α^i is a map of long exact sequences in M , so the class of M for which it is an isomorphism is closed under extensions. Induct on $\#M$: reduce to M killed by a prime.
3. **Shapiro plus induced modules.** Choose L/K finite Galois with G_L acting trivially on M . Then M embeds in $\text{Ind}_{G_L}^{G_K}(M|_{G_L})$, and for induced modules Shapiro's lemma turns the statement over K into the statement over L — where the module is **trivial**, i.e. Section 3.

That is the whole architecture. Serre, *Galois Cohomology* II.5; Milne, *Arithmetic Duality Theorems* I.2; Neukirch–Schmidt–Wingberg VII.2. There is also a conceptual repackaging: G_K is a duality group of cohomological dimension 2 with dualizing module the roots of unity, so the theorem is Poincaré duality for $\text{Spec } K$ viewed as a 2-dimensional object.

5. The numerical half: Tate's Euler characteristic formula

The companion result, proved by the same dévissage, is

$$(\#H^0(K, M) \cdot \#H^2(K, M)) / \#H^1(K, M) = \|\#M\|_K.$$

This is what upgrades “isotropic” to “**maximal** isotropic”, and it is the fact the other documents lean on.

Take $M = E[n]$, self-dual under the Weil pairing, so $\#H^2 = \#H^0 = \#E[n](K)$. Then

$$\#H^1(K, E[n]) = \#E[n](K)^2 \cdot |n|_K^{-2}, \quad \text{while} \quad \#(E(K)/n) = \#E[n](K) \cdot |n|_K^{-1}.$$

Hence $\#W^2 = \#H^1$: the local Kummer image $W = E(K)/nE(K)$ is **exactly** half-dimensional. Isotropy itself comes from the diagram $0 \rightarrow E(K)/n \rightarrow H^1(K, E[n]) \rightarrow H^1(K, E)[n] \rightarrow 0$: two Kummer classes pair through the image in $H^1(K, E)$, which is zero on $E(K)/n$. So W is **Lagrangian**.

The factor $|n|_K^{-1}$ is invisible away from n and is the entire subtlety at $v \mid n$. Checked for $n = 2$ at $v = 2$, where it is live:

curve	$\dim E[2](\mathbb{Q}_2)$	$\dim W_2$	$\dim H^1$
$y^2 + y = x^3 - x$ (37)	0	1	2
$y^2 + xy + y = x^3 + x^2 - x$ (89)	1	2	4
$y^2 + y = x^3 + x^2$ (43)	0	1	2
$y^2 = x^3 - 2x$	1	2	4
$y^2 = x^3 - x$	2	3	6

$\dim W_2 = \dim E[2](\mathbb{Q}_2) + 1$ against $\dim H^1 = 2 \dim E[2](\mathbb{Q}_2) + 2$: half throughout, with the +1 and +2 being exactly the $|2|_2^{-1}$ correction that is absent at odd places.

6. Symmetric versus alternating self-duality

Worth separating, because the two are easy to conflate and the difference is what makes the Kummer surface behave differently from the Pell conic.

- $\mathbb{Z}/2$ is self-dual **symmetrically**. Over $\mathbb{Q}_2$ the Hilbert form on $\mathbb{Q}_2^\times / (\mathbb{Q}_2^\times)^2 \cong \mathbb{F}_2^3$ is non-degenerate on an **odd**-dimensional space, so a Lagrangian would need dimension $3/2$ — there is none. The diagonal $(a, a)_2$ is non-trivial for $a \in \{-1, -2, -5, -10\}$, and the largest totally isotropic subspace has dimension 1 (e.g. spanned by 2).

- $E[n]$ is self-dual **alternatingly**, via the Weil pairing, on an **even**-dimensional space — and there Lagrangians exist. W_v is one.

That is the whole reason for the collapse in `density-bridge.typ` §2. On the Kummer surface the reciprocity sum pairs **two points**, both in the Lagrangian W_v , so it vanishes identically and gives $0 = 0$; the twisted pairing exists to break exactly this. The quaternion argument of `integral-bm.typ` pairs a point against a **class**, and $\mathbb{Z}/2$ has no Lagrangian to collapse into — so no such degeneration occurs there.

7. The tame case is soft, the wild case is not

For $\ell \neq p$ the module is unramified at good reduction, and $H_{\text{ur}(K,M)}^1$ and $H_{\text{ur}(K,M^D)}^1$ are exact annihilators of each other, each of order $\#H^0$. That alone gives the duality in the tame range, with no fundamental class — and it is precisely the “unramified isotropy” that kills β_v at good places throughout the survey.

p	unramified classes	order	half of $\#H^1$?
3, 5, 7, 11, 13	$\{1, u\}$	2	yes
2	$\{1, 5\}$	2	no — $\#H^1 = 8$

At $\ell = p$ the module is no longer unramified — the formal group is in the way — the count fails, and one is thrown back on `inv` and `LCFT` in full. Which is the same tame/wild line as `wild-symbols.typ`: soft and residue-field-sized away from the residue characteristic, genuinely analytic at it.

8. From finite modules to abelian varieties

From $0 \rightarrow E(K)/n \rightarrow H^1(K, E[n]) \rightarrow H^1(K, E)[n] \rightarrow 0$ and self-duality of $E[n]$, the two outer terms are exact annihilators of each other under the duality pairing, hence perfectly paired. Now take the inverse limit over n on the left, giving $E(K)^\wedge$, and the direct limit over n on the right, giving all of $H^1(K, E)$ (which is torsion). That is

$$E(K)^\wedge \times H^1(K, E) \longrightarrow \mathbb{Q}/\mathbb{Z},$$

the pairing used in `ec-density-bm.typ`. For a general abelian variety it is $A(K) \times H^1(K, \hat{A})$; for elliptic curves $\hat{E} = E$ by the principal polarisation, which is exactly why $E[n]$ is self-dual and why the Weil pairing appears where it does.

Companion file: `local-duality.gp`, run as

```
gp -q -s 4000000000 local-duality.gp < /dev/null > results/local-duality.txt
```

It verifies the $\mathbb{Z}/2$ case as non-degeneracy of the Hilbert symbol (exhaustive Gram matrices at $p = 2, 3, 5, 7, 11, 13$), the unramified annihilation and its failure at 2, the Lagrangian count for $E[2]$ over $\mathbb{Q}_2$ including the $|2|_2^{-1}$ correction, and the largest totally isotropic subspace of $\mathbb{Q}_2^\times / \text{sq}$.